

SECTION A: MULTIPLE CHOICE QUESTIONS - 10 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. START WITH SECTION A AND HAND THE OPTIC READER FORM IN AFTER 50 MINUTES.
2. First circle your answers on this paper and only mark the answers on the form when you are certain of your choice. You may not erase on the optic reader form.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use SIDE 2 of the optic reader form.
Fill in your personal information in pencil on side 2.
Fill in your student number from top to bottom and then code it.
6. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 10 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. BEGIN MET AFDELING A EN HANDIG DIE MERKLEESVORM IN NA 50 MINUTE.
2. Omkring jou antwoorde eers in hierdie vraestel en merk die antwoorde op die merkleesvorm eers as jy seker is van jou keuse. Jy mag nie uitvee op die merkleesvorm nie.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik KANT 2 van die merkleesvorm.
Vul u persoonlike inligting in potlood in op kant 2.
Vul u studentnommer van bo na onder in en kodeer dit.
6. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

$$\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right) =$$

(Notation / Notasie: $\cos^{-1} = \arccos = \text{bgcos}$)

(a)	$-\frac{2\pi}{3}$	(b)	$-\frac{\pi}{3}$	☒ (c)	$\frac{\pi}{3}$
(d)	$\frac{2\pi}{3}$	(e)	π	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

The domain of $f(x) = \sin^{-1}(2x + 1)$ is / Die definisieversameling van $f(x) = \sin^{-1}(2x + 1)$ is

(Notation / Notasie: $\sin^{-1} = \arcsin = \text{bgsin}$)

(a)	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	(b)	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	(c)	$(-1, 1)$
☒ (d)	$[-1, 0]$	(e)	$(-1, 0)$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

$$\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x \sin(x + 1)} =$$

(a)	0	(b)	∞	(c)	-1
(d)	1	(e)	Does not exist / Bestaan nie	(f)	None of these / Geen van hierdie

Question 4 / Vraag 4

$$\lim_{x \rightarrow \infty} \tan^{-1} \frac{6}{x} =$$

(a)	0	(b)	∞	(c)	$-\frac{\pi}{2}$
(d)	1	(e)	$\frac{\pi}{2}$	(f)	None of these / Geen van hierdie

Question 5 / Vraag 5

Let / Laat $f(x) = \begin{cases} x^2 - 1 & \text{if / indien } x \neq 1 \\ 4 & \text{if / indien } x = 1 \end{cases}$

Which one of the following statements, is true?

- I. $\lim_{x \rightarrow 1} f(x)$ exists
- II. $f(1)$ exists
- III. f is continuous at $x = 1$

Watter een van die volgende bewerings, is waar?

- I. $\lim_{x \rightarrow 1} f(x)$ bestaan
- II. $f(1)$ bestaan
- III. f is kontinu by $x = 1$

(a)	Only / Slegs I	(b)	Only / Slegs II	(c)	Only / Slegs III
(d)	I and / en II	(e)	All of them / Almal van hierdie	(f)	None of these / Geen van hierdie

Question 6 / Vraag 6

The horizontal asymptotes (if any) of $f(x) = \frac{ax^3}{b + cx + dx^2}$ is given by:

Die horisontale asimptote (indien enige) van $f(x) = \frac{ax^3}{b + cx + dx^2}$ word gegee deur:

(a)	$y = \frac{a}{b}$	(b)	$y = 0$
(c)	$y = \frac{a}{d}$	(d)	$y = a$
(e)	There are no horizontal asymptotes Daar is geen horisontale asimptote	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

$$\lim_{\theta \rightarrow 0} \frac{\sin 3\theta}{\theta + \tan \theta} =$$

(a)	0	(b)	$\frac{1}{2}$	(c)	$\frac{3}{2}$
(d)	1	(e)	Does not exist / Bestaan nie	(f)	None of these / Geen van hierdie

Question 8 / Vraag 8

Given the function $g(x) = e^x - 2x$. For which value(s) of x will the function have a horizontal tangent?

Gegee die funksie $g(x) = e^x - 2x$. Vir watter waarde(s) van x het die funksie 'n horisontale raaklyn?

(a)	2	(b)	e^2	(c)	$\ln 2$
(d)	2.5	(e)	None of these / Geen van hierdie		

Question 9 / Vraag 9

Given that / Gegee dat $f(1) = -1$ and / en $f'(1) = 4$.

What is the equation of the tangent line to f in the point where $x = 1$?

Wat is die vergelyking van die raaklyn aan f in die punt $x = 1$?

(a)	$y = 4x - 3$	(b)	$y = 4x - 5$	(c)	$y = -x + 5$
(d)	$y = -4x + 5$	(e)	$y = -x - 3$	(f)	None of these / Geen van hierdie

Question 10 / Vraag 10

$$\lim_{x \rightarrow -\infty} \frac{3x^2 + 27}{x^3 - 27} =$$

(a)	$-\infty$	(b)	∞	(c)	0
(d)	-9	(e)	Does not exist / Bestaan nie	(f)	None of these / Geen van hierdie

SECTION B: 35 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. Do not use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
If you need more space for an answer, use the facing page and indicate it clearly by framing it.
5. Show all steps and calculations and explain your answers.

AFDELING B: 35 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. Korreger-vloeistof ("Tipp-Ex") mag nie gebruik word nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
5. Toon alle stappe en berekeninge en verduidelik jou antwoorde.

Question 1 / Vraag 1

Fill in the answer only in the space provided. Rough work can be done on the opposing page and will not be marked.

Vul slegs die antwoord in die toegekende spasie in. Berekeninge kan op die teenblad gedoen word en sal nie nagesien word nie.

Given / Gegee $f(x) = \frac{2x^2 - 7x - 15}{x^2 - x - 20}$.

1.1 f has vertical asymptote(s) given by:

f het vertikale asimptoot(e) gegee deur:

$$x = -4$$

1.2 f has horizontal asymptote(s) given by:

f het horisontale asimptoot(e) gegee deur:

$$y = 2$$

Question 2 / Vraag 2

Fill in the answer only in the space provided. Rough work can be done on the opposing page and will not be marked.

Vul slegs die antwoord in die toegekende spasie in. Berekeninge kan op die teenblad gedoen word en sal nie nagesien word nie.

2.1 Given the functions $h(x) = (x+1)^2$, $f(x) = 4x$ and $g(x)$, where $f(x) \leq g(x) \leq h(x)$ for all $x \in \mathbb{R}$.

Gegee die funksies $h(x) = (x+1)^2$, $f(x) = 4x$ en $g(x)$, waar $f(x) \leq g(x) \leq h(x)$ vir alle $x \in \mathbb{R}$.

$$\lim_{x \rightarrow 1} g(x) =$$

4

1

2.2 Sketch a function F that satisfies all of the following requirements:

Skets 'n funksie F wat aan al die volgende voorwaardes voldoen:

I. F has a jump discontinuity at $x = 2$

I. F het 'n sprong diskontinuiteit in $x = 2$

II. F is right continuous in $x = 2$

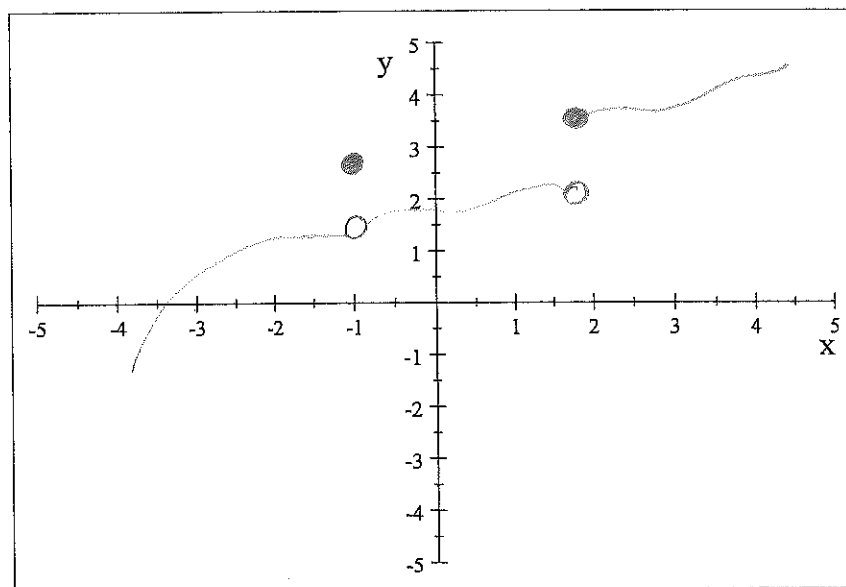
II. F is regs kontinu in $x = 2$

III. F has a removable discontinuity at $x = -1$

III. F het 'n verwyderbare diskontinuiteit in $x = -1$

IV. $F(-1)$ is defined

IV. $F(-1)$ is gedefinieer



2

2.3 $G(x) = e^{3+x} + x^e + \pi x = e^3 e^x + x^e + \pi x$

$$G'(x) = e^3 e^x + e x^{e-1} + \pi$$

$$G''(x) = e^3 e^x + e(e-1)x^{e-2}$$

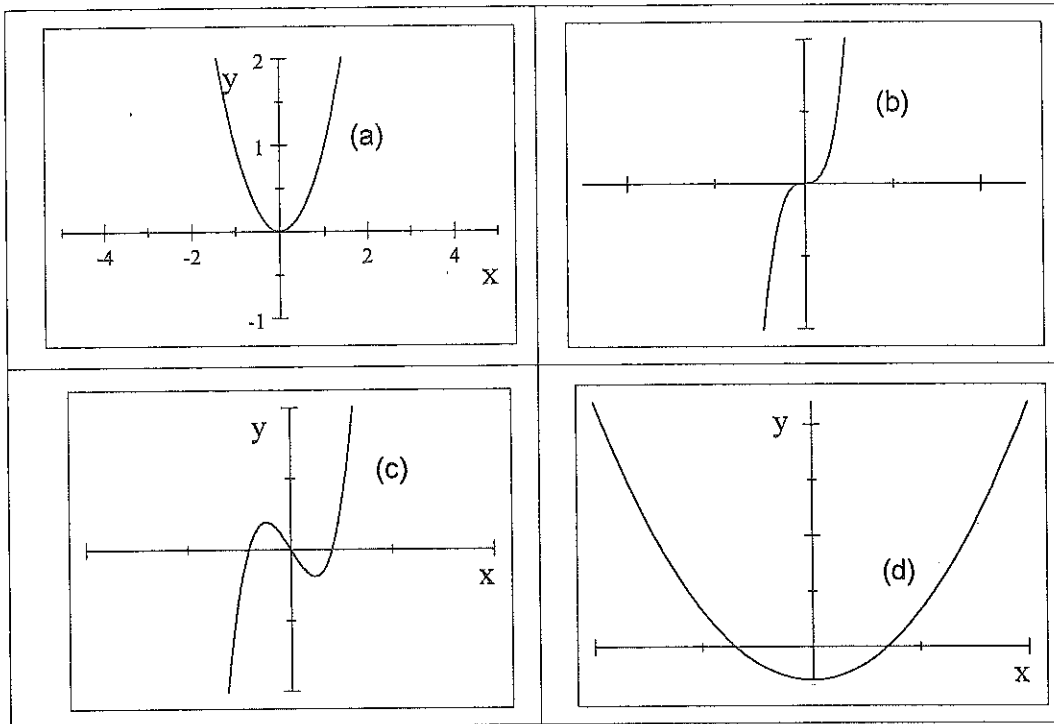
1

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Question 3 / Vraag 3

The figure shows graphs of f, f', f'' and f''' .

Die figuur vertoon die grafieke van f, f', f'' en f''' .



Identify each curve by inserting a, b, c, or d in the appropriate space.

Identifiseer elke kromme deur a, b, c of d in te vul in die toegekende spasie.

f is given by graph / word gegee deur grafiek

f' is given by graph / word gegee deur grafiek

f'' is given by graph / word gegee deur grafiek

f''' is given by graph / word gegee deur grafiek

(c)
(d)
(b)
(a)

Question 4 / Vraag 4

Find the value of / Vind die waarde van

$$\sin\left[2\cos^{-1}\left(-\frac{2}{3}\right)\right]$$

(Notation / Notasie: $\cos^{-1} = \arccos = \text{bgcos}$)

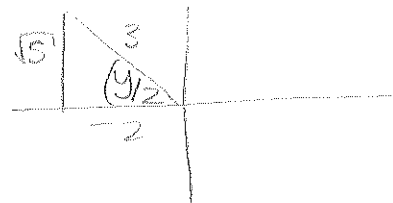
Let $y = \cos^{-1}\left(-\frac{2}{3}\right) \Rightarrow \cos y = -\frac{2}{3}$ and $y \in [0, \pi]$

$$\begin{aligned} \text{So } \sin 2y &= 2 \sin y \cos y \\ &= 2 \left(\frac{\sqrt{5}}{3}\right) \left(-\frac{2}{3}\right) = -\frac{4\sqrt{5}}{9} \end{aligned}$$



Let $y = 2\cos^{-1}\left(-\frac{2}{3}\right) \Rightarrow \cos\left(\frac{y}{2}\right) = -\frac{2}{3}$ and $\frac{y}{2} \in [0, \pi]$

$$\begin{aligned} \sin(y) &= 2 \sin\left(\frac{y}{2}\right) \cos\left(\frac{y}{2}\right) \\ &= 2 \left(\frac{\sqrt{5}}{3}\right) \left(-\frac{2}{3}\right) = -\frac{4\sqrt{5}}{9} \end{aligned}$$



3

Question 5 / Vraag 5

Calculate the following limits, if it exists.

Bepaal die volgende limiete, indien dit bestaan.

5.1 $\lim_{x \rightarrow 2^+} \frac{2+5x-3x^2}{|2-x|}$

$$\begin{aligned} |2-x| &= \begin{cases} 2-x & \text{if } 2-x > 0 \\ -(2-x) & \text{if } 2-x < 0 \end{cases} \\ &= \begin{cases} 2-x & \text{if } x < 2 \\ -(2-x) & \text{if } x > 2 \end{cases} \end{aligned}$$

$$= \lim_{x \rightarrow 2^+} \frac{(2-x)(1+3x)}{-(2-x)}$$

$$= \lim_{x \rightarrow 2^+} -(1+3x)$$

$$= -7$$

2

~~X~~

5.2 $\lim_{x \rightarrow 0} \sqrt{x} \tan^{-1}\left(\frac{1}{x}\right)$ (Notation / notation: $\tan^{-1} = \arctan = \text{bgtan}$)

$$-\frac{\pi}{2} < \tan^{-1}\left(\frac{1}{x}\right) < \frac{\pi}{2}$$

$$-\frac{\pi}{2}\sqrt{x} < \sqrt{x} \tan^{-1}\left(\frac{1}{x}\right) < \frac{\pi}{2}\sqrt{x}$$

$$\lim_{x \rightarrow 0} -\frac{\pi}{2}\sqrt{x} = 0, \quad \lim_{x \rightarrow 0} \frac{\pi}{2}\sqrt{x} = 0$$

$\Rightarrow$ By the squeeze theorem:

$$\lim_{x \rightarrow 0} \sqrt{x} \tan^{-1}\left(\frac{1}{x}\right) = 0$$

5.3 $\lim_{x \rightarrow -\infty} \frac{2x+5}{\sqrt{x^2+4}} = \lim_{x \rightarrow -\infty} \frac{x(2+\frac{5}{x})}{-x\sqrt{1+\frac{4}{x^2}}} = -2$

~~2~~ 3

Question 6 / Vraag 6

6.1 Complete the definition: A function f is continuous at $x = a$ if

Voltooi die definisie: 'n Funksie f is kontinu in $x = a$ as

$$\lim_{x \rightarrow a} f(x) = f(a)$$

$$\begin{aligned} \lim_{x \rightarrow a^-} f(x) &= \\ \lim_{x \rightarrow a^+} f(x) &= \\ f(a) & \end{aligned}$$

1

6.2 For which value(s) of a and b will the function f be continuous at $x = 4$?

Clearly show how you use the definition in Question 6.1.

Vir watter(s) van a en b sal die funksie f kontinu wees in $x = 4$?

Toon duidelik aan hoe u die definisie in Vraag 6.1 gebruik.

$$f(x) = \begin{cases} ax + (5 - 2a) & \text{if / as } x < 4 \\ 6 & \text{if / as } x = 4 \\ b(x - 6)^2 + 3 & \text{if / as } x > 4 \end{cases}$$

$$\bullet \lim_{x \rightarrow 4^-} ax + (5 - 2a) = 4a + 5 - 2a = 2a + 5$$

$$\bullet \lim_{x \rightarrow 4^+} b(x - 6)^2 + 3 = 4b + 3$$

$$\bullet f(4) = 6$$

$$\Rightarrow 2a + 5 = 6 \quad \text{and} \quad 4b + 3 = 6$$

$$\Rightarrow a = \frac{1}{2} \quad \text{and} \quad b = \frac{3}{4}$$

3

Question 7 / Vraag 7

7.1 Complete the definition: f is differentiable in $x = a$ if

Voltooi die definisie: f is differensieerbaar in $x = a$ as

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists.}$$

1

7.2 Use the limit definition to determine $f'(4)$ if $f(x) = \sqrt{x+2}$.

Gebruik die limietdefinisie en bepaal $f'(4)$ indien $f(x) = \sqrt{x+2}$.

$$\begin{aligned} f'(4) &= \lim_{x \rightarrow 4} \frac{\sqrt{x+2} - \sqrt{6}}{x-4} \times \frac{\sqrt{x+2} + \sqrt{6}}{\sqrt{x+2} + \sqrt{6}} \\ &= \lim_{x \rightarrow 4} \frac{(x+2) - 6}{(x-4)(\sqrt{x+2} + \sqrt{6})} \\ &= \lim_{x \rightarrow 4} \frac{x-4}{(x-4)(\sqrt{x+2} + \sqrt{6})} \\ &= \frac{1}{2\sqrt{6}} \end{aligned}$$

3

$$\begin{aligned} f'(4) &= \lim_{h \rightarrow 0} \frac{\sqrt{h+6} - \sqrt{6}}{h} \times \frac{\sqrt{h+6} + \sqrt{6}}{\sqrt{h+6} + \sqrt{6}} \\ &= \lim_{h \rightarrow 0} \frac{h+6-6}{h(\sqrt{h+6} + \sqrt{6})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{h+6} + \sqrt{6})} \\ &= \frac{1}{2\sqrt{6}} \end{aligned}$$

Question 8 / Vraag 8

8.1 Prove that if f is differentiable at $x = a$ then f is continuous at $x = a$.

Bewys dat as f differensieerbaar is by $x = a$ dan is f kontinu by $x = a$.

P.T.P : $\lim_{x \rightarrow a} f(x) = f(a)$

Given : $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists.

Proof : $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x) - f(a) + f(a)]$
 $= \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} \times (x - a) \right] + \lim_{x \rightarrow a} f(a)$
 $= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times \lim_{x \rightarrow a} (x - a) + f(a)$
 $= f'(a) \times 0 + f(a)$
 $= f(a)$ □

8.2 Show, by example, that the converse of this theorem is not true.

Gee 'n teenvoorbeeld om aan te toon dat die omgekeerde van die stelling nie waar is nie.

$f(x) = |x|$ (or others) is
a continuous function but not
differentiable.

3

1

Question 9 / Vraag 9

Determine the equation of the parabola, $y = ax^2 + bx$, if $y = 3x - 2$ is the equation of the tangent line at $(1, 1)$.

Bepaal die vergelyking van die parabool, $y = ax^2 + bx$, as $y = 3x - 2$ die vergelyking van die raaklyn by $(1, 1)$ is.

• Slope of tangent

$$y' = 2ax + b = 3$$

$$y'(1) = 2a + b = 3 \quad \text{--- (1)}$$

• $ax^2 + bx = 3x - 2$ in $(1, 1)$

$$\Rightarrow a + b = 3 - 2$$

$$\Rightarrow a + b = 1 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad a = 2$$

$$\Rightarrow b = -1$$

Equation of parabola is

$$y = 2x^2 - x$$